Secretary Problem Research

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Lecture 8: Partially Adversarial SP with Examination Tue 31 May 2022 08:47

1 Partially Adversarial SP

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1.1 Problem formulation

Random arrival. n items are totally ordered. A random permutation σ is taken uniformly from the permutation group S_n. The classical SP corresponds to finding a stopping time that most likely stops on $t = \sigma(1)$. We will always use numbers to denote the true rank of items, so the incoming stream of items is represented by $\sigma(1, \dots, n)$.

Adversarial noise. The adversary adds a random noise with resultant permutation π. This permutation is not over the arriving order of object, but over the observed rank, i.e. $\pi(j) = i$ means that the item with actual rank j will be found by the player to be i-th best, if the player observes all items without examining them.

Restriction on adversary. To restrict the power of adversary, we demand π to be long-range stable, i.e. $\pi(i) < \pi(j)$ whenever $i < j$ and $j - i > c$, where the constant c is known to the algorithm.

Examination. For each incoming item the player can choose from three options: Accept, $\mathsf{Decline}$, and $\mathsf{Examine}$. If the player chooses to $\mathsf{Examine}$, the current item will be added to a set E of examinated items, and the next item will be forfeited and left unobserved. We use E_t to denote the items examined by the start of t-th round. The $\mathsf{Examine}$ option is not available if the current item is the last item.

Observable quantities. Let O_t be the observed total order on round t. It is a total order on $[t]$ induced by $\pi\sigma^{-1}$. Let $\widehat{O}_t$ be the actual total order on round t. It is a total order on $[t]$ induced by σ^{-1}. Let P_t be the total order obtained from examination, defined by restricting $\widehat{O}_t$ to E_t. Note that O_t and P_t may not be compatible since O_t is subject to observational "mistakes" caused by the noise.

It is equivalent, but sometimes more convenient, to use the notion of relative rank instead of total order. We define $\mathrm{RR}_t(i) = \sum_{s=1}^{t} \mathbf{1}\left(\pi\sigma^{-1}(s) \geq \pi\sigma^{-1}(i)\right)$ to be the observed relative rank function.

Decision rule. Let $(\mathcal{F}_t)_{t\in[n]}$ be the sequence of σ-algebras with each $\mathcal{F}_t$ generated by the random variables $O_1 \times P_1, \dots, O_t \times P_t$, that is,

$$\mathcal{F}_t = \sigma\left(O_1 \times P_1, \dots, O_t \times P_t\right) = \sigma\left(O_t \times P_t\right).$$

By a stopping time, we mean a random variable τ taking values in $[n]$ and satisfying the property $\{\tau = t\} \in \mathcal{F}_t$. The decision rule is a stopping time τ together with a sequence of indicators $\mathsf{Exm}_t := \mathbf{1}(t \in E_{t+1})$ such that

$$\mathsf{Exm}_{t+1} \in \mathcal{F}_t \text{ and } \mathsf{Exm}_t\mathsf{Exm}_{t+1} = 0 \text{ and } \mathbf{1}\left(\tau = t\right)\mathsf{Exm}_t = 0.$$

Goal. Our goal is to choose a decision rule (τ, Exm) to maximize $\Pr\left[\tau = \sigma(1)\right]$.

1.2 Preliminaries

We list several obvious facts about this problem construction.

Lemma 1.
1. The relative ranks behave the same: $\mathrm{RR}(t)$ is uniformly sampled from $1, \ldots, t+1$ and independent from each other.
2. When π do not depend on σ, $\sigma\pi$ and σ have exactly same distribution.

1.3 Case without examination

We first do not allow examination, and see how far the algorithm can get.

1.3.1 Weak adversarial, do not know anything

Consider the weakest adversary, that can only pick a single π without knowing about the algorithm. The competitive ratio is

$$\begin{aligned}
&\inf_{\pi} \sup_{\tau} \mathbb{E}_{\sigma} \left[\Pr\left[\tau = \sigma\pi(1) \right] \right] \\
&= \inf_{k} \inf_{\pi:\pi(1)=k} \sup_{\tau} \mathbb{E}_{\sigma} \left[\Pr\left[\tau = \sigma(k) \right] \right] \\
&\leq \inf_{k} \sup_{\tau} \mathbb{E}_{\sigma} \left[\Pr\left[\tau = \sigma(k) \right] \right].
\end{aligned}$$

The problem reduces to a generalized post-doc variant of SP, where we want the item with absolute rank k instead of 1. Each time k is provided, we may specifically design our algorithm to maximize the probability of choosing the rank k item. This is significantly harder to solve as compared to the classical SP, in its value function is no longer monotone: the function

$$\Pr\left[\mathrm{AR}(t) = m \mid \mathrm{RR}(t) = r \right] = \frac{\binom{m-1}{r-1}\binom{n-m}{t-r}}{\binom{n}{t}}$$

is not monotone wrt r for any fixed m. No explicit solution has been found so far. We can show that the worst case when $c < \frac{n}{2}$ is attained at $\pi(1) = c$, and at $\pi(1) = \left\lfloor \frac{n}{2} \right\rfloor$ when $c \geq \frac{n}{2}$.

1.3.2 Weak adversarial, know algorithm

Now we assume that the adversary adapts to each algorithm, but cannot depend on the random permutation σ. The competitive ratio is now

$$\sup_{\tau} \inf_{\pi} \mathbb{E}_{\sigma} \left[\Pr\left[\tau = \sigma\pi(1) \right] \right]$$

Note that the generalized postdoc variant of SP is an upper bound to the problem:

$$\begin{aligned}
&= \sup_{\tau} \inf_{k} \inf_{\pi:\pi(1)=k} \mathbb{E}_{\sigma} \Pr\left[\tau = \sigma(k) \right] \\
&\leq \sup_{\tau} \inf_{k} \mathbb{E}_{\sigma} \Pr\left[\tau = \sigma(k) \right].
\end{aligned}$$

This time we need to tackle all sub-problems using only one algorithm. In other words, the algorithm's probability to choose the 1, 2, $\ldots$, c-th item are compared, and the minimum one is the CR.

For lower bound, we consider the following algorithm: When each item arrives, if $\mathrm{RR}(t) > c$, reject the item. Otherwise, accept with probability p, where p is a tuning parameter. Now for some algorithm τ and fixed k, We derive analytic result for this algorithm. Its performance monotonously decrease wrt $\pi(1)$ for fixed n and p, and always approach the trivial lower bound $\frac{1}{n}$.

Consider another algorithm: Let the first pn items to go by, and then select the first item with relative rank better or equal to c. This algorithm makes no distinction among the first c items, so it suffices to analyze its behavior on the classic SP.

1.3.3 Strong adversarial, knows τ and σ

$$\sup_{\tau} \mathbb{E}_{\sigma} \inf_{\pi} \left[\Pr\left[\tau = \sigma\pi(1)\right]\right].$$

1.3.4 Strong adversarial, knows τ, σ and r

$$\sup_{\tau} \mathbb{E}_{\sigma} \left[\mathbb{E}_{\mathbf{r}} \left[\inf_{\pi} \mathbf{1}\left(\tau = \sigma\pi(1)\right)\right]\right].$$

In this case the adversary knows the random seed used by the algorithm, so it's able to make hindsight decisions. In this setting, the algorithm achieve zero cr. This can be shown by calling to an auxiliary problem of adversarial best-choice within the top c items, where no deterministic algorithm can guarantee nonzero competitive ratio against an adversary.

1.4 Lower Bound for an algorithm

Algorithm: examine each item with $\mathrm{RR}_t(t) \leq c$. Accept the current item if $t \geq T$ and the item is the best so far.

It is equivalent to view this game as follows: when an item is examined, the last available item is omitted, i.e. the omitted item is taken from the end of the queue. To simplify the analysis, we also allow the omitted item to appear as an ordinary item. The player wins only if the best itme is chosen, AND the item is not omitted.

$$\begin{aligned}
\Pr[\text{win}] &= \Pr\left[\text{win CSP with } n \text{ items and } t^* + |E_{t^*}| \leq n\right] \\
&\geq \max_{a \leq n} \Pr\left[\text{win CSP with } n \text{ items and } t^* + |E_{t^*}| \leq n \cap t^* \leq a\right] \\
&= \max_{a \leq n} \Pr\left[\text{win CSP with } n \text{ items } |t^* + |E_{t^*}| \leq n \cap t^* \leq a\right] \Pr\left[t^* + |E_{t^*}| \leq n | t^* \leq a\right] \Pr\left[t^* \leq a\right] \\
&= \max_{a \leq n} \Pr\left[\text{win CSP with } a \text{ items } |t^* + |E_{t^*}| \leq n \cap t^* \leq a\right] \Pr\left[|E_{t^*}| \leq n - a\right] \Pr\left[t^* \leq a\right] \\
&= \max_{a \leq n} \Pr\left[\text{win CSP with } a \text{ items }\right] \Pr\left[|E_{t^*}| \leq n - a\right] \Pr\left[t^* \leq a\right] \\
&\geq \max_{a \leq n} \frac{1}{e} \Pr\left[|E_a| \leq n - a\right] \Pr\left[t^* \leq a\right].
\end{aligned}$$

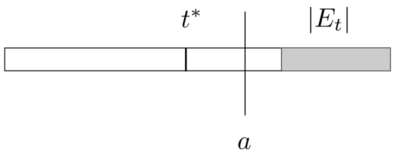

Figure 1: LowerBound

For $t \geq c$ we have

$$|E_t| = c - 1 + c\sum_{s=c}^{t} B_s$$

where $B_s \sim \text{Ber}(\frac{1}{s})$.

Using Hoeffding's Bound:

$$\Pr\left[\sum_{c}^{t} B_s - \sum_{c}^{t}\frac{1}{s} \leq \beta\right] \geq 1 - \exp\left(-\frac{2\beta^2}{t-c+1}\right)$$
$$\Pr\left[|E_t| \leq c(\beta + \sum_{s=c}^{t}\frac{1}{s}) + c - 1\right] \geq 1 - \exp\left(-\frac{2\beta^2}{t-c+1}\right).$$

Fix some a. Let β be the greatest real number such that

$$c(\beta + \sum_{s=c}^{a}\frac{1}{s}) + c - 1 \leq n - a.$$

Therefore,

$$\beta = \frac{n-a}{c} - 1 - H(a) + H(c-1)$$

where we use $H(a)$ to denote the a-th harmonic number.

Applying this bound, we now have

$$\Pr[\text{win}] \geq \max_{a \leq n} \frac{1}{e}\frac{a}{n}\left(1 - \exp\left(-\frac{2\beta^2}{a-c+1}\right)\right).$$

Take $c = 2$ for an example. When $a = 70$, we have the numerical bound $\Pr[\text{win}\ ] \geq 0.674\frac{1}{e}$, which is a non-trivial improvement from the examine-everything algorithm which guarantees only a competitive ratio of $0.5\frac{1}{e}$.

[[Linear Algebra]]